

An Internship Report
Of
Data Engineering Virtual Internship

Submitted To

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IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree
of

B.Tech (Computer Science and Engineering)



By

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Batch: 2024-2028

CANDIDATE'S DECLARATION

I, PRATYUSH KHATRI, do hereby declare that the internship report titled “Data Engineering with Python Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

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ACKNOWLEDGEMENT

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I express my gratitude to my parents for their blessings.

Signature:

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3. INTERNSHIP COMPLETION CERTIFICATE



This is to certify that
Pratyush Khatri

has successfully completed the

Freecodecamp: Data Engineering with Python and AI course

We commend their dedication and expertise in the field.

Certificate ID: 6a940a9209b7ae6ed40a49fa

30 Aug 2026
Date of issue

A Brudaru
Adrian Brudaru
Co-founder & CDO

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the Virtual Internship undertaken in the domain of **Data Engineering with Python and AI/LLMs**, conducted by freeCodeCamp and dltHub. The program was completed as part of the academic requirements for the Bachelor of Technology in Computer Science and Engineering at IILM University, Greater Noida.

The primary aim of this internship was to move beyond simple, custom scripts to platform-engineering resilience and reliability in modern data ingestion pipelines. The internship followed a flexible, hands-on learning structure where core data engineering concepts—such as REST API extraction, dynamic schema evolution, data contracts, DuckDB loading, incremental loads (SCD2), destination data lakes/warehouses, and orchestration—were learned through comprehensive YouTube video tutorials and hands-on Google Colab code exercises. Upon working through the tutorial modules and practical code implementations, technical understanding was formally validated through a comprehensive assessment quiz covering the entire curriculum. On successfully clearing the quiz, an official Master of ELT Best Practices certification was awarded by dltHub and freeCodeCamp.

4.2 Organization Profile

- **dltHub:** The core maintainer of `dlt` (Data Load Tool), an open-source Python library designed to automate the complex, repetitive aspects of data engineering—such as schema detection, nested JSON flattening, state tracking, and incremental loading.
- **freeCodeCamp:** A global non-profit organization providing open-access technical education, hands-on programming pathways, and recognized certifications for software engineers worldwide.

4.3 Problem Statement

Traditional ETL scripts suffer from high maintenance overhead, frequent breakage due to API schema drift, complex nested JSON processing, and memory management issues when scaling batch extraction. Developers frequently spend significant time writing custom boilerplate code to handle authentication, pagination, rate limits, manual table creation, dynamic schema updates, and

state management for incremental loads. The problem addressed in this internship was to build automated, self-healing ELT data pipelines that handle real-world API complexities, dynamic schema normalization, data governance contracts, target storage loading, and deployment using modern tools and AI assistance.

4.4 Project Objectives

- Understand data engineering fundamentals, comparing batch and real-time streaming architectures.
- Extract data efficiently from REST APIs, handling authentication, pagination, rate limits, and memory safety.
- Automate dynamic schema evolution and flatten nested JSON structures into relational forms.
- Implement **Data Contracts** (evolve, freeze, discard) and schema change alerting systems (e.g., Slack webhooks).
- Master write dispositions (append, replace, merge) and implement incremental loading along with Slowly Changing Dimensions (SCD2).
- Load structured data into in-memory databases (DuckDB), data warehouses (BigQuery, Snowflake, Redshift), and data lakehouses (AWS S3, Parquet, Apache Iceberg).
- Deploy and orchestrate data pipelines using GitHub Actions, Crontab, Dagster, and Apache Airflow.
- To successfully clear the Final Credential Validation / Internship Grade Point Assessment.

4.5 Scope of the Project

The scope spans the full modern ELT lifecycle: from raw REST API extraction, custom Python generators, and SQL-to-SQL copying to normalization using `dlt`, loading into DuckDB, cloud data warehouses, and S3-based lakehouse formats. It encompasses setting up automated workflow orchestration scripts, state management, backfilling, and leveraging LLMs to design and audit code. Learning was executed within interactive Google Colab environments provided by `dltHub` and `freeCodeCamp`.

4.6 Technologies and Tools Used

- **Languages & Core Libraries:** Python, Requests, Pandas, PyArrow, ConnectorX
- **Ingestion & ELT Framework:** `dlt` (Data Load Tool) / `dltHub`

- **Databases & Warehouses:** DuckDB, MySQL, Google BigQuery, Snowflake, Amazon Redshift
- **Data Lakes & Formats:** AWS S3, Parquet, Apache Iceberg, Delta Lake, AWS Glue Catalog, AWS Athena
- **Orchestration & Deployment:** GitHub Actions, Cron, Dagster, Apache Airflow
- **AI & Development Tools:** OpenAI GPT models, Cursor IDE, JetBrains PyCharm, Google Colab

4.7 System Architecture

The system architecture follows a modular, 5-layer pipeline model:

1. **Data Sources Layer:** Ingests raw data from REST APIs, SQL databases, and external raw files.
2. **dlt Extraction Layer:** Manages streaming chunks, credentials (`.dlt/secrets.toml`), schema evolution, and data contract enforcement.
3. **Transformation & Normalization Layer:** Automatically flattens nested JSON, handles SQL-to-SQL copying, state tracking, and SCD2 historic tracking.
4. **Destination Layer:** Loads processed data into OLAP engines (DuckDB), cloud data warehouses, or lakehouse storage formats (AWS S3/Iceberg).
5. **Orchestration & AI Layer:** Triggers automated runs via GitHub Actions, Cron, Dagster, or Airflow, with pipeline templates generated using LLM prompts and Cursor IDE.

4.8 Methodology

The methodology followed a structured course outline covering foundational theory, Colab coding tutorials, practical implementations, and a final certification assessment.

Topic / Module	Key Technical Focus & Description
1. Introduction & Foundations	Core data engineering concepts, moving from scripts to platform engineering.
2. Batch vs. Streaming Processing	Comparing batch processing with real-time stream processing architectures.
3. Extracting Data from REST APIs	Pagination, authentication, rate limits, and memory-safe extraction.
4. Data Normalization & Schemas	Flattening nested JSON arrays and managing dynamic schema evolution.

5. Loading Data into DuckDB	Converting Python data structures into columnar DuckDB tables.
6. Introduction to dlt	Configuring <code>dlt</code> resources, secret management, and automated normalization.
7. Data Contracts & Alerting	Enforcing contract modes (evolve, freeze, discard) and alerting via Slack hooks.
8. Write Dispositions & Incremental	Append, replace, cursor tracking, state persistence, backfilling, and SCD2.
9. Performance Tuning	Parallel workers, async requests, and memory management optimizations.
10. Lakes, Lakehouses & Warehouses	Parquet, Iceberg, Delta Lake, AWS S3, Glue Catalogs, and BigQuery loading.
11. Deployment & Orchestration	Deploying pipelines via GitHub Actions, Cron, Dagster, and Apache Airflow.
12. Building Pipelines with AI/LLMs	Designing LLM prompts, generating docs, and vibe-coding using Cursor.

4.9 Expected Outcomes

- Ability to build scalable, self-healing Python ELT data pipelines using `dlt`.
- Elimination of custom JSON flattening boilerplate via automated dynamic normalization.
- Enforcement of robust data governance using Data Contracts and dynamic schema drift alerts.
- Proficiency in configuring stateful incremental loads, backfilling, and Slowly Changing Dimension (SCD Type 2) tracking.
- Hands-on experience deploying data to cloud data warehouses, lakehouses, and modern orchestration tools.
- Practical skill in utilizing AI/LLM assistants to design, document, and build production data pipelines.

4.10 Certificate of Completion

The Internship Completion Certificate issued by dltHub and freeCodeCamp, confirming successful completion of the **Freecodecamp: Data Engineering with Python and AI** course and the award of the **Master of ELT Best Practices** credential, is attached in Chapter 3 of this report. The official certificate serves as primary verification of course completion and skill validation.

REFERENCES/ BIBLIOGRAPHY

- dltHub & freeCodeCamp, "Certificate of Completion – Master of ELT Best Practices," Certificate ID: 6a940a9209b7ae6ed40a49fa, 2026.
- dltHub Education Repository, "Data Engineering with dlt (freeCodeCamp Course Code & Notebooks)," https://github.com/dlt-hub/dlthub-education/tree/main/courses/freecodecamp/de_with_dlt_2025
- freeCodeCamp & dltHub, "Data Engineering with Python and AI Colab Notebooks," https://colab.research.google.com/drive/1nQt6G6gNyIlJuKi_X-RMwbUyYDdkPI9k